Name:- Vaishnavi rajesh bomnale

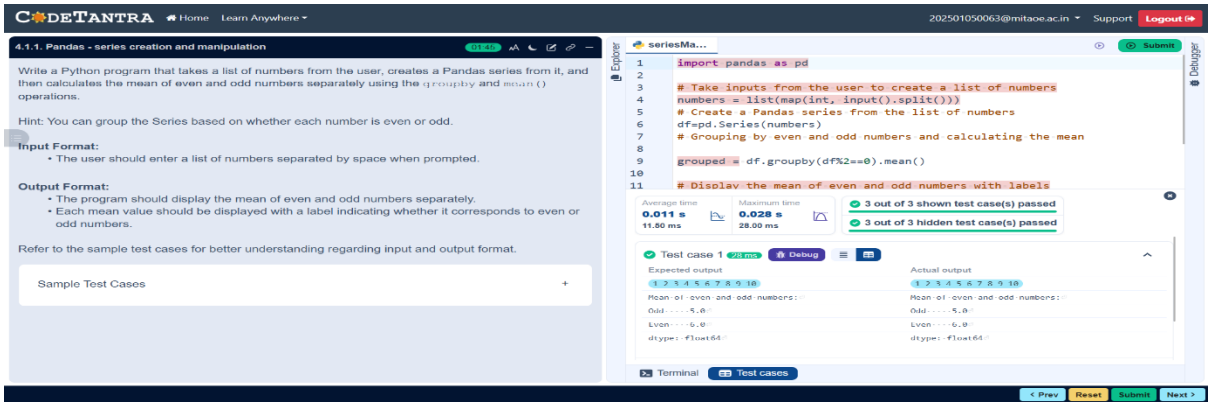
PRN:-202501050063

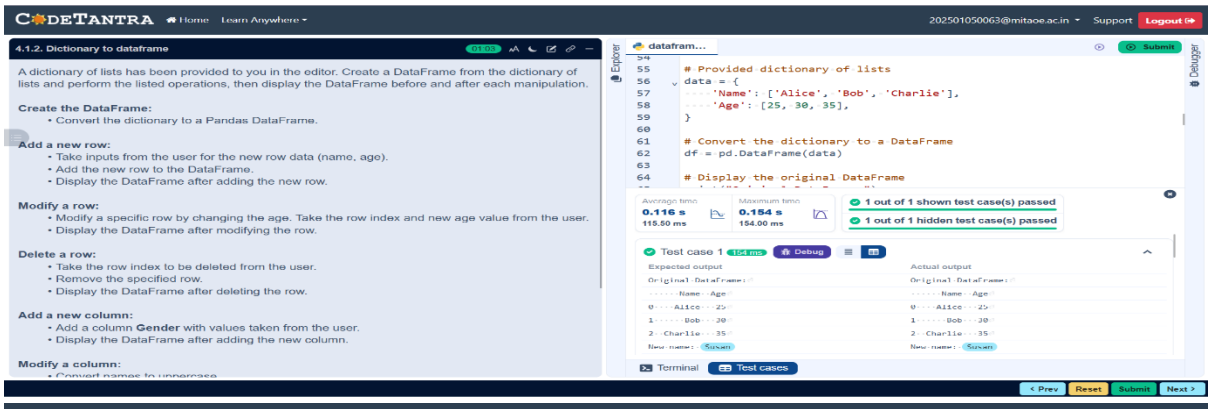
Batch :- CH3

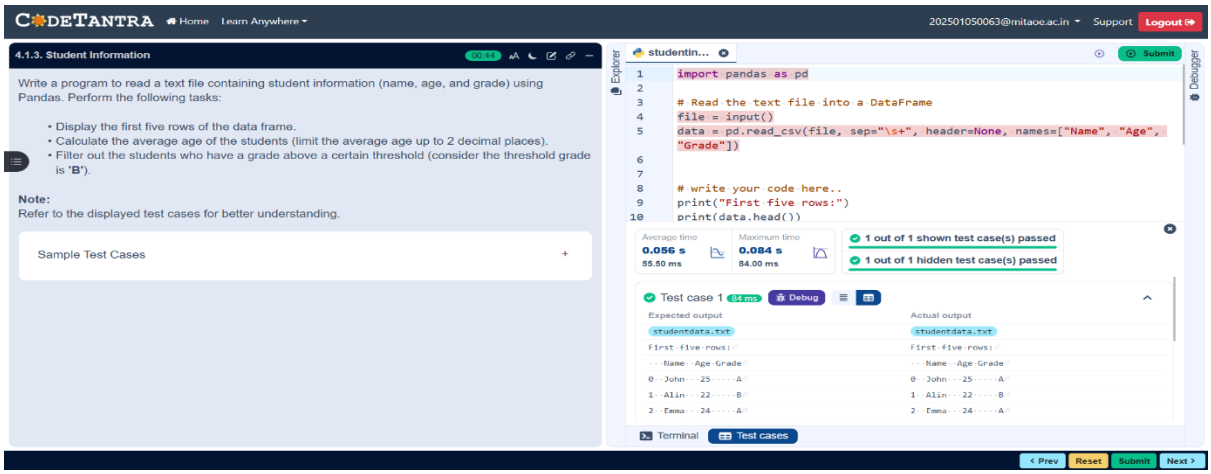
Sub :- EDS

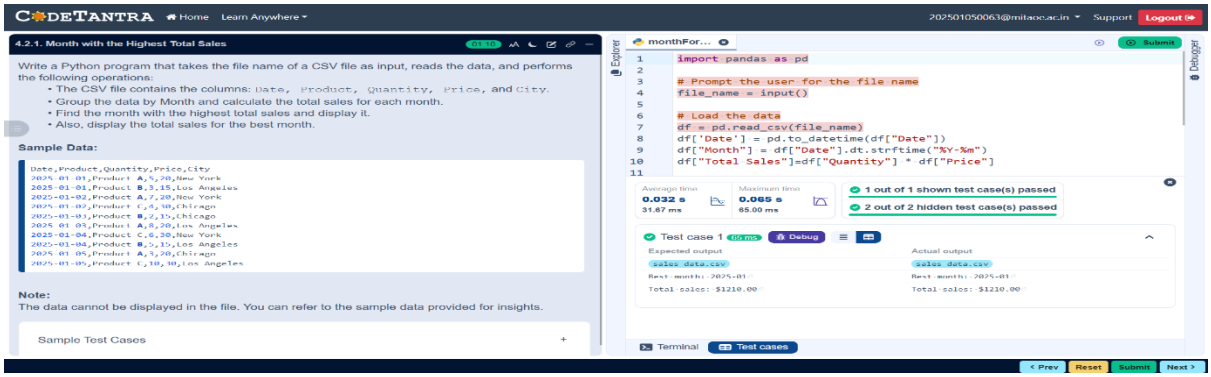
Screenshot of practical no 4:

Practical lab assignment:





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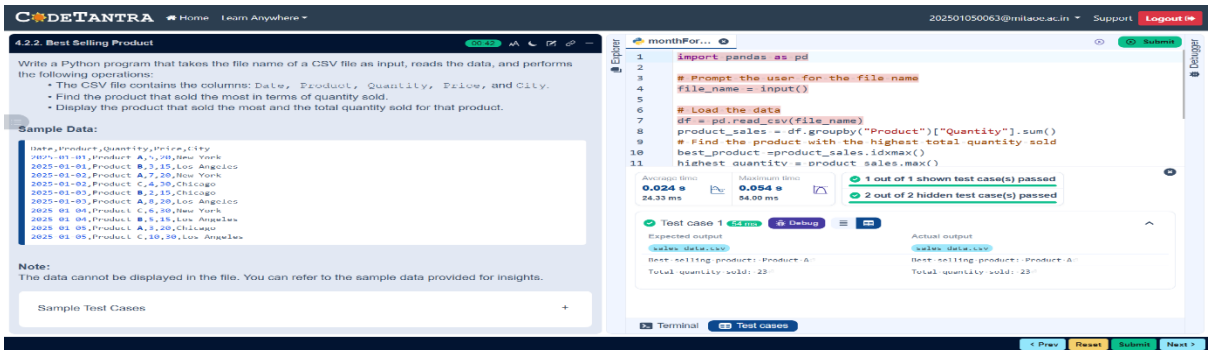
4.2.1. Month with the Highest Total Sales
01:10
Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:
• The CSV file contains the columns: Date, Product, Quantity, Price, and City.
• Group the data by Month and calculate the total sales for each month.
• Find the month with the highest total sales and display it.
• Also, display the total sales for the best month.
Sample Data:
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.
Sample Test Cases
monthFor...
Submit
Explorer
Debugger
import pandas as pd
# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
df['Date'] = pd.to_datetime(df["Date"])
df["Month"] = df["Date"].dt.strftime("%Y-%m")
df["Total Sales"]=df["Quantity"] * df["Price"]
Average time
0.032 s
31.67 ms
Maximum time
0.085 s
85.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
Debug
Expected output
sales_data.csv
Best month: 2025-01
Total sales: $1210.00
Actual output
sales_data.csv
Best month: 2025-01
Total sales: $1210.00
Terminal
Test cases
< Prev
Reset
Submit
Next >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4.2.2. Best Selling Product
00:42
Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:
• The CSV file contains the columns: Date, Product, Quantity, Price, and City.
• Find the product that sold the most in terms of quantity sold.
• Display the product that sold the most and the total quantity sold for that product.
Sample Data:
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.
Sample Test Cases
monthFor...
Submit
Explorer
Debugger
import pandas as pd
# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
product_sales = df.groupby("Product")["Quantity"].sum()
# Find the product with the highest total quantity sold
best_product =product_sales.idxmax()
highest_quantity = product_sales.max()
Average time
0.024 s
24.33 ms
Maximum time
0.054 s
54.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
24 ms
Debug
Expected output
sales_data.csv
Best selling product: Product A
Total quantity sold: 23
Actual output
sales_data.csv
Best selling product: Product A
Total quantity sold: 23
Terminal
Test cases
< Prev
Reset
Submit
Next >


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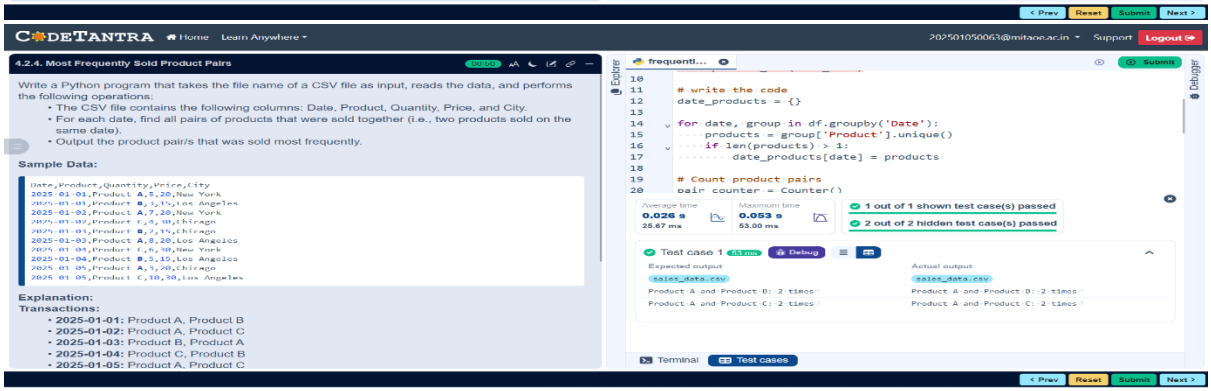
4.2.3. City that Sold the Most Products
00:40
Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:
• The CSV file contains the columns: Date, Product, Quantity, Price, and City.
• Group the data by City and calculate the total quantity of products sold for each city.
• Find the city that sold the most products (based on the total quantity sold).
Sample Data:
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.
Sample Test Cases
monthFor...
Submit
Explorer
Debugger
import pandas as pd
# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
# write the code..
city_sales = df.groupby("City")["Quantity"].sum()
best_city=city_sales.idxmax()
Average time
0.024 s
24.00 ms
Maximum time
0.049 s
49.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
49 ms
Debug
Expected output
sales_data.csv
City sold the most products: Los Angeles
Actual output
sales_data.csv
City sold the most products: Los Angeles
Terminal
Test cases
< Prev
Reset
Submit
Next >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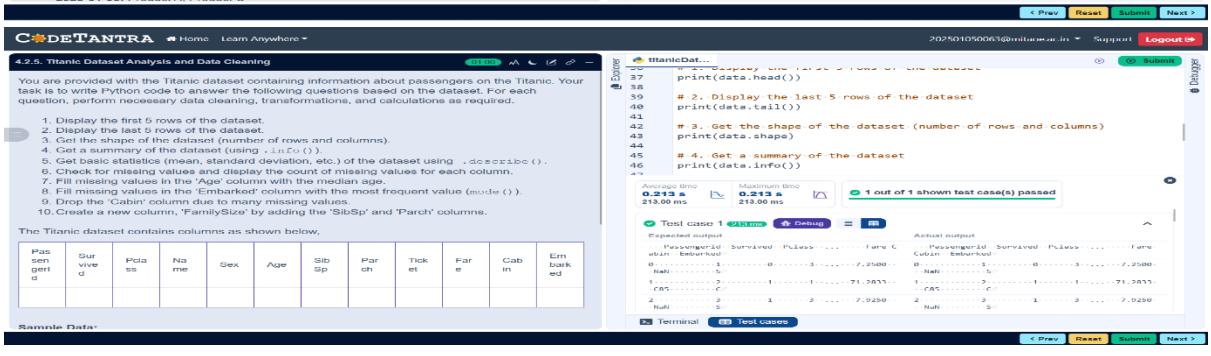
4.2.4. Most Frequently Sold Product Pairs
00:00
Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:
• The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
• For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
• Output the product pair/s that was sold most frequently.
Sample Data:
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
Explanation:
Transactions:
• 2025-01-01: Product A, Product B
• 2025-01-02: Product A, Product C
• 2025-01-03: Product B, Product A
• 2025-01-04: Product C, Product B
• 2025-01-05: Product A, Product C
frequentI...
Submit
Explorer
Debugger
# write the code
date_products = {}
for date, group in df.groupby('Date'):
products = group['Product'].unique()
if len(products) > 1:
date_products[date] = products
# Count product pairs
pair_counter = Counter()
Average time
0.028 s
25.67 ms
Maximum time
0.053 s
53.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
53 ms
Debug
Expected output
sales_data.csv
Product A and Product B: 2 times
Product A and Product C: 2 times
Actual output
sales_data.csv
Product A and Product B: 2 times
Product A and Product C: 2 times
Terminal
Test cases
< Prev
Reset
Submit
Next >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4.2.5. Titanic Dataset Analysis and Data Cleaning
01:00
You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.
1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using .info()).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (mode()).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.
The Titanic dataset contains columns as shown below,
Passengerid
Survived
Pclass
Name
Sex
Age
SibSp
Parch
Ticket
Fare
Cabin
Embarked
Sample Data:
titanicDat...
Submit
Explorer
Debugger
print(data.head())
# 2. Display the last 5 rows of the dataset
print(data.tail())
# 3. Get the shape of the dataset (number of rows and columns)
print(data.shape)
# 4. Get a summary of the dataset
print(data.info())
Average time
0.213 s
213.00 ms
Maximum time
0.213 s
213.00 ms
1 out of 1 shown test case(s) passed
Test case 1
213 ms
Debug
Terminal
Test cases
< Prev
Reset
Submit
Next >

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

01:22

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Passengerld	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

Explorer titanicDat... Submit Debugger

```
print(data.groupby('Pclass')['Survived'].mean())

#2. Calculate the survival rate by embarked location (Embarked_S)

print(data.groupby('Embarked_S')['Survived'].mean())

#3. Calculate the survival rate by family size

print(data.groupby('FamilySize')['Survived'].mean())

#4. Calculate the survival rate by being alone
```

Average time 0.109 s 109.00 ms | Maximum time 0.109 s 109.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 109 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S

Terminal Test cases

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

00:51

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

Passengerld	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

Explorer titanicDat... Submit Debugger

```
# 5. Percentage of children (Age < 18) who survived
children = data[data['Age'] < 18]
children_survival_rate = children['Survived'].mean()
print(children_survival_rate)

# 6. Percentage of adults (Age >= 18) who survived
adults = data[data['Age'] >= 18]
adults_survival_rate = adults['Survived'].mean()
print(adults_survival_rate)

# 7. Median age of survivors
```

Average time 0.081 s 81.00 ms | Maximum time 0.081 s 81.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 81 ms Debug

Expected output	Actual output
female 233	female 233
male 109	male 109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male 468	male 468
female 81	female 81
Name: Sex, dtype: int64	Name: Sex, dtype: int64

Terminal Test cases

Prev Reset Submit Next